

Guideline/Protocol Title	Guidelines for the Management of Skin and Soft Tissue Infections in Pediatrics
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Committee Review	Antimicrobial Stewardship Subcommittee Pharmacy and Therapeutics Committee
Target Population	Pediatric Patients with Skin and Soft Tissue Infections
Overview	This guideline provides evidence-based management of skin and soft tissue infections
Effective Date	11/2022
Revised Date	N/A
Expiration Date	11/2025
Schedule for Periodic Review	Every 3 years
Implementation Strategy	See below. Each service will be educated on the new guideline.
Education Strategy	ID pharmacist, ID physician, and ASC service line representatives will educate each affected service. Email communication will be sent out as well.
Information Technology Needs	Access to CareWeb, UK Antimicrobial Stewardship App and Pharmacy SharePoint

Guideline for the Management of Skin and Soft Tissue Infections in Pediatrics

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GUIDELINES FOR THE MANAGEMENT OF SKIN AND SOFT TISSUE INFECTIONS IN PEDIATRICS

PURPOSE OF GUIDELINES: To provide evidence-based guidelines for the treatment of skin and soft tissue infections at the University of Kentucky Health Care

- GOALS:**
- Early diagnosis and treatment
 - Evidence based use of antibiotics
 - Prompt referral/consult for severe infections including necrotizing fasciitis
-

BACKGROUND: Skin and soft tissue infections (SSTI) are the second most common type of infection leading to hospitalization in the United States (US).⁽⁶⁾ In 2014 the Infectious Diseases Society of America (IDSA) released an updated practice guideline for the diagnosis and management of skin and soft tissue infections (SSTIs).⁽⁷⁾ These guidelines were created to improve patient outcomes and use of health care resources while limiting the unintended consequences of unnecessary antibiotic use.^(8, 9) An institutional guideline has been developed to adapt the IDSA guidelines and existing literature into practice for the management of SSTIs. Please note that infections involving bones and joints are not addressed in this guideline.

TARGET POPULATION: Pediatric patients with SSTIs

DIAGNOSIS AND INITIAL WORKUP OF SKIN AND SOFT TISSUE INFECTIONS:

Diagnosis of SSTIs are predominantly clinical with features including erythema, warmth, edema, and pain among others. They can be generally placed into two categories of purulent (furuncles, carbuncles, abscess) and non-purulent (erysipelas, cellulitis, necrotizing fasciitis, myonecrosis) and further divided into subcategories of mild, moderate and severe based upon degree of systemic involvement. There are other SSTI classification schemes proposed based upon patient presentations that have been used in other countries but none yet approved or widely adopted in the US to date.⁽¹⁰⁾

For purulent SSTIs, the mainstay of treatment typically includes incision and drainage (with gram stain and culture when feasible) and adjunctive antibiotics for moderate and severe cases. For small or already draining collections, the use of warm compresses alone may be sufficient. For non-purulent SSTIs, the mainstay of treatment is antibiotics, with emergent or urgent surgical debridement (with gram stain and cultures) in select moderate and severe cases. Generally, empiric antibiotics, blood cultures, and cultures of tissue or aspirate are recommended in patients who are septic, immunocompromised, have a malignancy, or have unusual predisposing risk factors including but not limited to immersion injury, animal bites, neutropenia or severe cell-mediated immunodeficiency. Consider early infectious disease consultation in patients with bacteremia, anticipation of greater than two weeks of anti-infective treatment, immunocompromising conditions, history of multi-drug resistant organisms, or recurrence of infection.

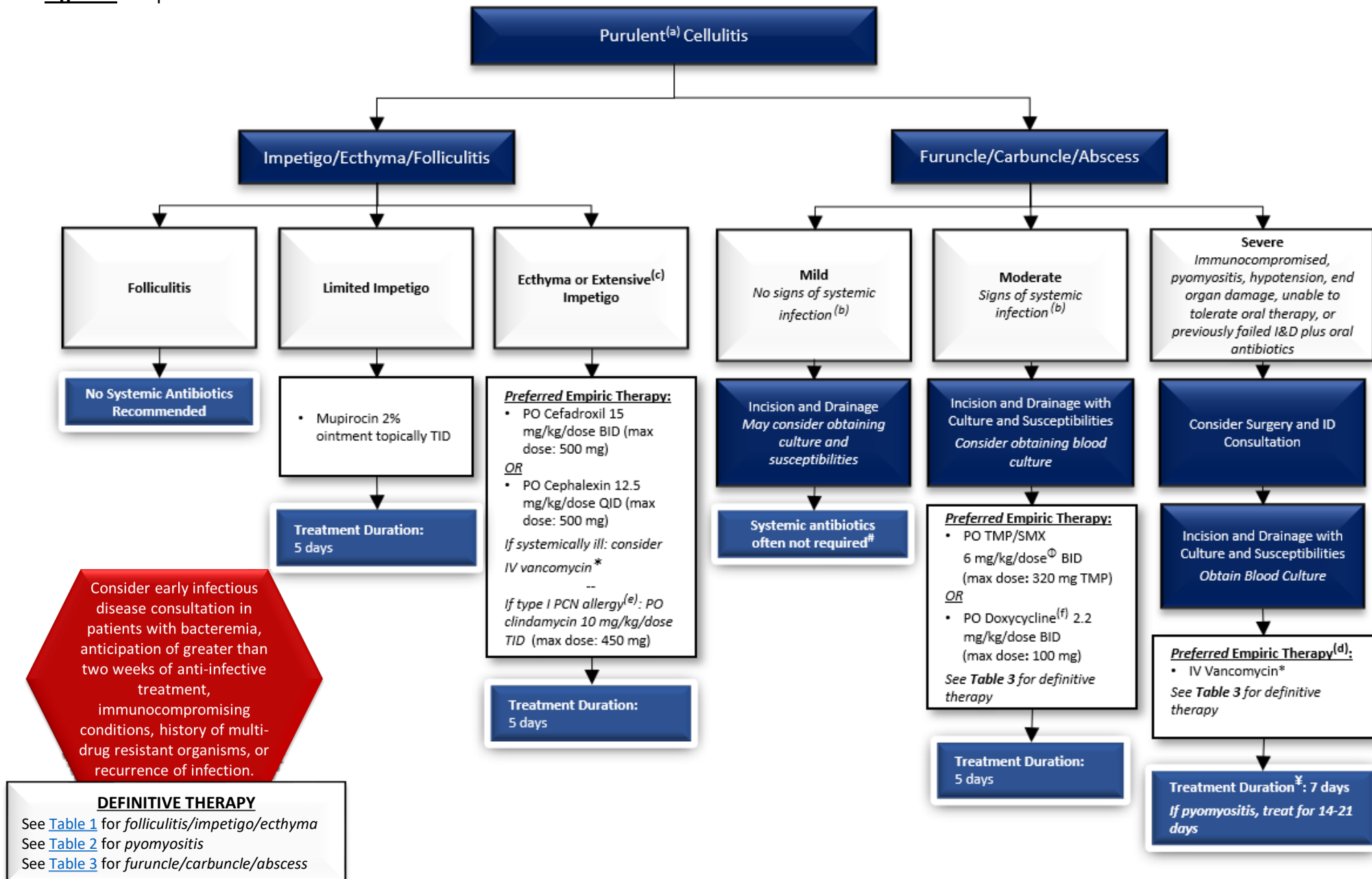
PENICILLIN ALLERGIES:

Assessment of beta-lactam allergies should include the type of reaction, date of occurrence, treatment measure taken, and any antibiotics tolerated since the reaction occurred. Cross-reactivity between penicillin and first-generation cephalosporins with similar R-side chains in patients with type I reactions (anaphylaxis, angioedema, or urticaria) is estimated at only 1%, based on recent studies.⁽⁵⁾

SUSCEPTIBILITIES AND LOCAL RESISTANCE:

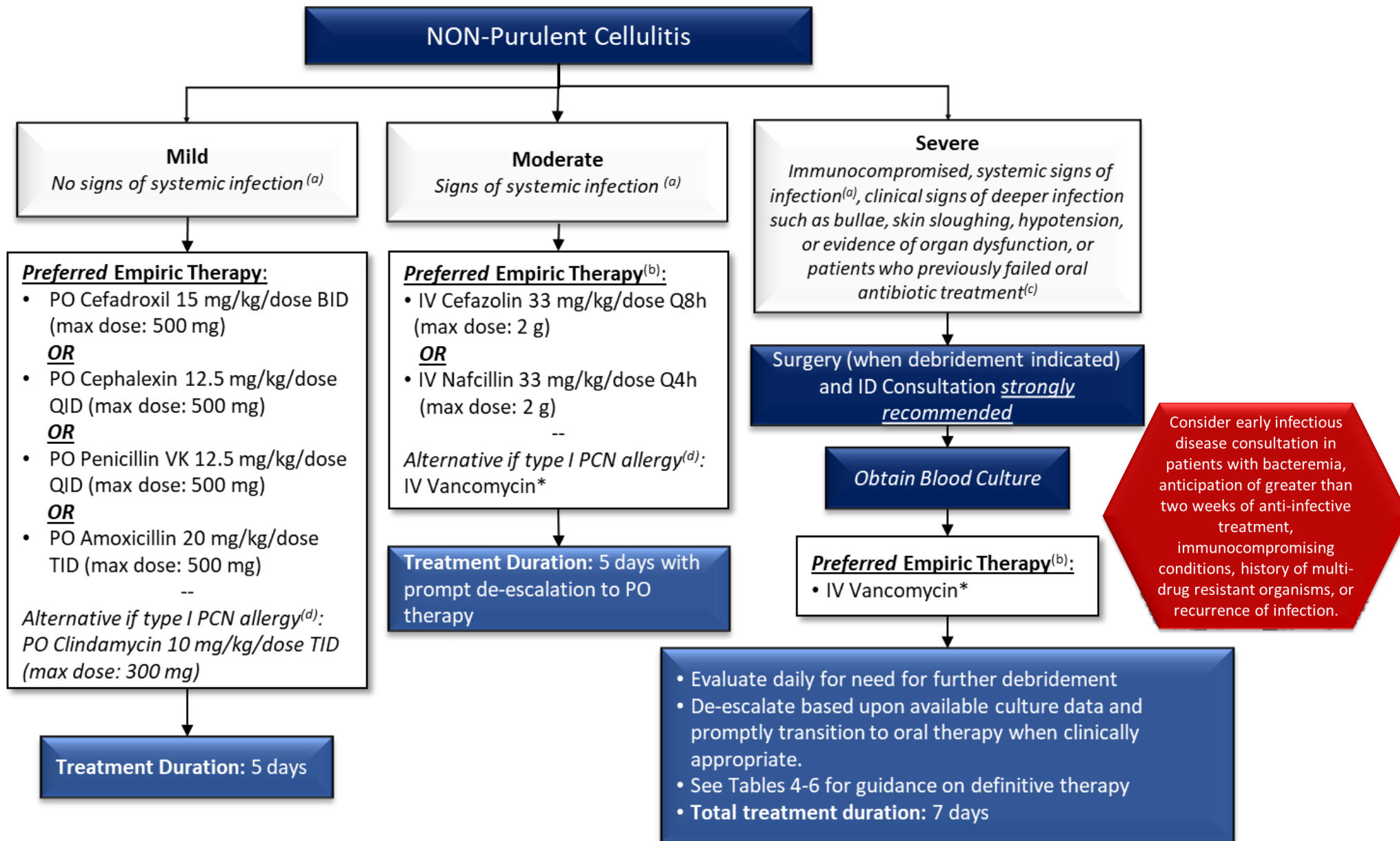
Local antibiogram data demonstrate that 20-30% of *S. aureus* (MRSA and MSSA) isolates are resistant to clindamycin. Trimethoprim/sulfamethoxazole, doxycycline, and vancomycin represent empiric agents with adequate activity (>90% susceptibility) against *S. aureus* isolates at Kentucky Children's Hospital (KCH).

Figure 1: Empiric Treatment of *Purulent* Cellulitis



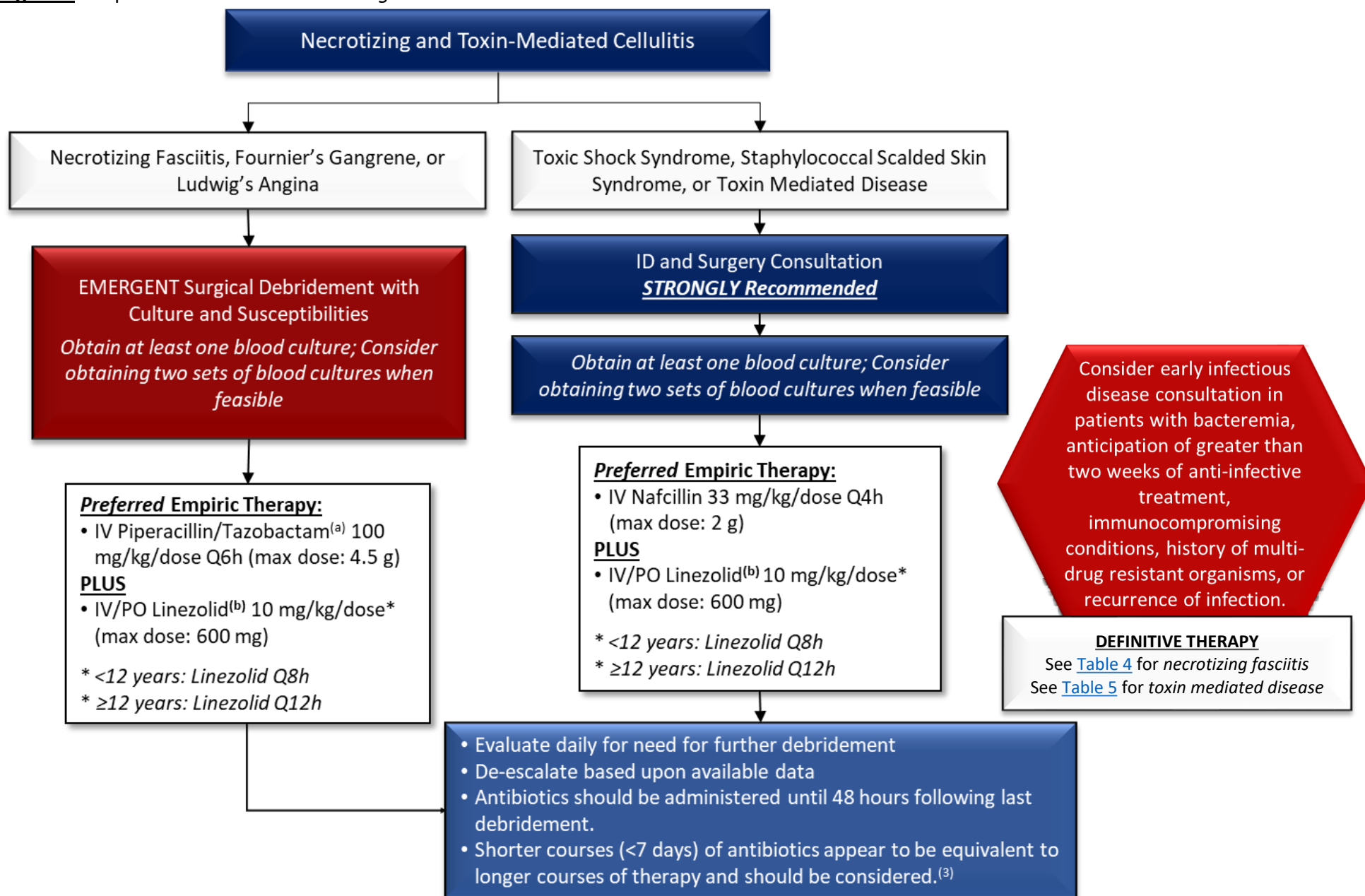
*Vancomycin dosed per PED-Vancomycin IV order set or pharmacy to dose; ¥: Treatment duration based on patient response and may vary based on ID consultant recommendations. For the treatment of necrotizing infections, shorter courses of antimicrobial therapy (<7 days) appears to be equivalent to longer therapy and should be considered⁽³⁾; Ⓞ: dosing of TMP/SMX based on trimethoprim component; #: May consider up to 5 days of PO TMP/SMX or doxycycline if abscess culture(s) positive for MRSA, patient is immunocompromised, or if there is surrounding cellulitis; (a): If purulent with surrounding cellulitis, follow purulent algorithm; (b): Signs of systemic infection include: fever (>38°C), tachycardia, tachypnea, leukocytosis, or leukopenia; (c): Impetigo is considered extensive when there are numerous lesions or in outbreaks affecting several people; (d): Consider conversion to PO therapy after favorable clinical response observed; (e): Type I reaction includes anaphylaxis, angioedema, or urticaria; Cross-reactivity between penicillins and first-generation cephalosporins with similar R-side chains is estimated at only 1%, based on recent studies.⁽⁵⁾ (f): Original data describing enamel staining are from studies with tetracycline. Recent comparative data in children suggest that doxycycline is NOT likely to cause visible teeth staining or enamel hypoplasia in children. The American Academy of Pediatrics recommend that doxycycline can be administered in any pediatric patient for durations ≤21 days without regard to the patient's age.⁽⁴⁾

Figure 2: Empiric Treatment of *NON-Purulent* Cellulitis



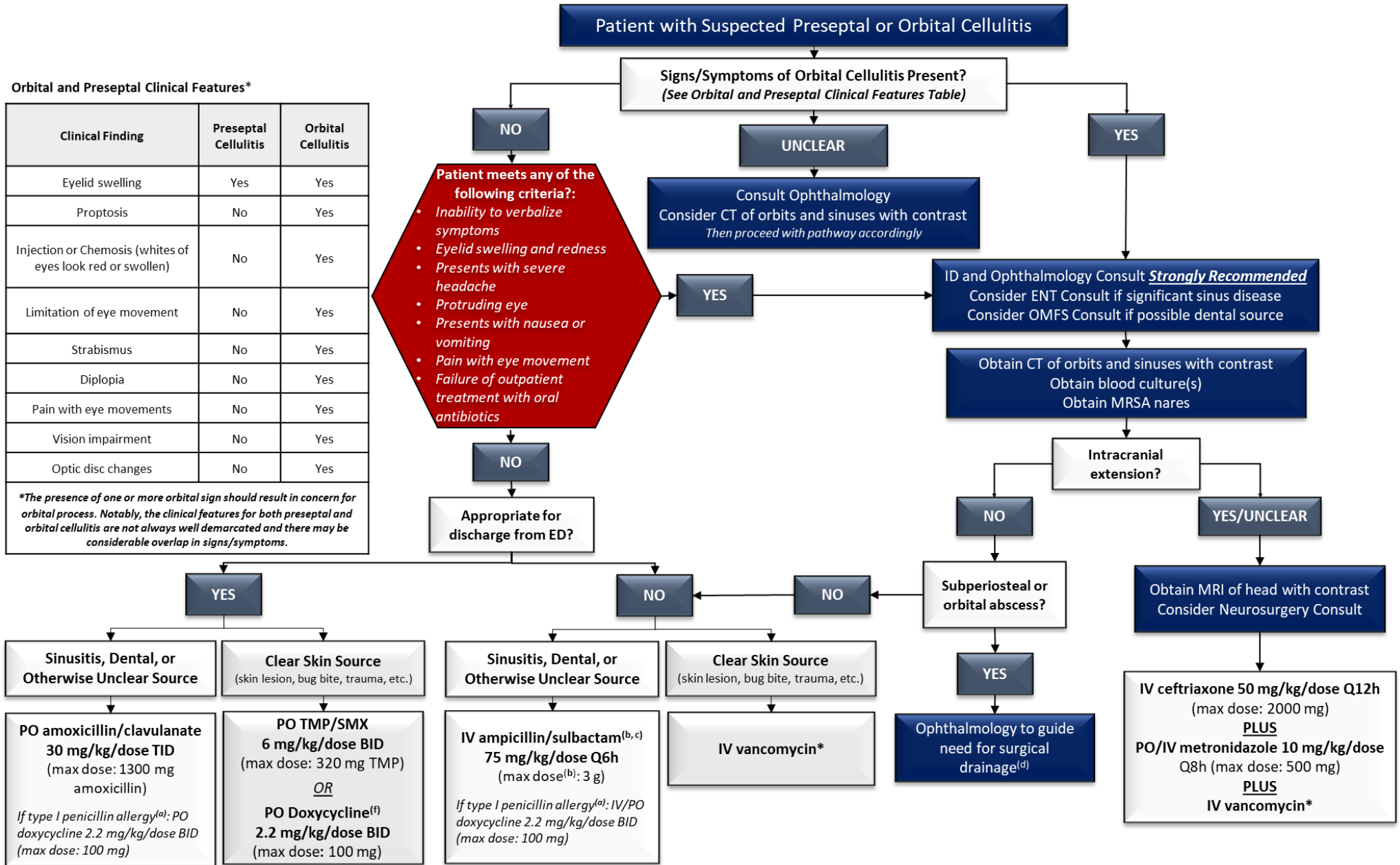
*Vancomycin dosed per pediatric IV vancomycin order set [PED-Vancomycin IV] or pharmacy to dose; **(a)**: Signs of systemic infection include: fever (>38°C), tachycardia, tachypnea, leukocytosis, or leukopenia; **(b)**: Consider conversion of IV to PO therapy after favorable clinical response observed and bacteremia ruled out; **(c)**: Failure of oral antibiotics defined as worsening of symptoms following ≥48-72 hours of appropriately selected and optimally dosed antistaphylococcal antibiotic therapy; **(d)**: Type I reaction includes anaphylaxis, angioedema, or urticaria; in patients with minor penicillin allergies (non-specific rashes, intolerance, etc.), cross reactivity with cephalosporins is low (<3%)^(1, 2)

Figure 3: Empiric Treatment of *Necrotizing and Toxin-Mediated Cellulitis*



(a): Piperacillin/tazobactam dosing is based on piperacillin component. Piperacillin/tazobactam 4.5 g contains 4000 mg piperacillin and 500 mg tazobactam; **(b):** Linezolid provides broad Gram-positive (including MRSA) coverage in addition to anti-toxin activity. Myelosuppression, namely thrombocytopenia and anemia, has been reported with linezolid and typically occurs with durations ≥14 days and is usually reversible with discontinuation of therapy. Patients on SSRIs or SNRIs should be monitored for signs/symptoms of serotonin syndrome if linezolid is initiated.

Figure 4: Management of Patients with Suspected Preseptal or Orbital Cellulitis



*Vancomycin dosed per pediatric IV vancomycin order set [PED-Vancomycin IV] or pharmacy to dose; **(a)**: Type I reaction includes anaphylaxis, angioedema, or urticaria; in patients with minor penicillin allergies (non-specific rashes, intolerance, etc.), cross reactivity with cephalosporins is low (<3%)^(1,2); **(b)**: Ampicillin/sulbactam dosing is based on ampicillin component; Ampicillin/sulbactam 3 g contains 2 g ampicillin and 1 g sulbactam. **(c)**: Patients admitted with orbital cellulitis typically warrant 72 hours of parenteral (IV) antibiotics prior to stepping down to oral therapy. **(d)**: The decision to pursue surgical drainage of subperiosteal abscess (SPA) based on Garcia/Harris criteria. Indications for surgical management include 1) Age of patient ≥ 9 years 2) Presence of frontal sinusitis 3) Nonmedial location of SPA 4) Large SPA 5) Suspicion of anaerobic subperiosteal infection (e.g., presence of gas within the abscess space as visualized on CT scan) 6) Recurrence of SPA after previous drainage 7) Evidence of chronic sinusitis (e.g., nasal polyps) 8) Acute optic nerve or retinal compromise 9) Infection of dental origin; **(f)**: Original data describing enamel staining are from studies with tetracycline. Recent comparative data in children suggest that doxycycline is NOT likely to cause visible teeth staining or enamel hypoplasia in children. The American Academy of Pediatrics recommend that doxycycline can be administered in any pediatric patient for durations ≤21 days without regard to the patient's age.⁽⁴⁾

Table 1: Treatment of Impetigo and Folliculitis

Impetigo and Folliculitis	
Classification	Recommendations and Treatment
Impetigo (Limited Disease)	- Mupirocin 2% ointment apply topically TID Treatment duration: 5 days
Impetigo (Extensive disease ^(a))	- PO cefadroxil 15 mg/kg/dose BID (max dose: 500 mg) <u>OR</u> - PO cephalexin 12.5 mg/kg/dose QID (max dose: 500 mg) Treatment duration: 5 days - If systemically ill, consider IV vancomycin* <i>Type I Penicillin allergy^(b):</i> Clindamycin^(c) 10 mg/kg/dose PO TID (max dose: 300 mg)
Ecthyma	
Folliculitis	- No systemic antibiotics needed - Consider use of warm compress

*Vancomycin dosed per pediatric IV vancomycin order set [PED-Vancomycin IV] or pharmacy to dose; **(a)**: Impetigo is considered extensive when there are numerous lesions or in outbreaks affecting several people; **(b)**: Type I reaction includes anaphylaxis, angioedema, or urticaria; in patients with minor penicillin allergies (non-specific rashes, intolerance, etc.), cross reactivity with cephalosporins is low (<3%)^(1,2) **(c)**: Clindamycin susceptibility rates per the 2021/2022 KCH antibiogram are only 69% for MRSA and 79% for MSSA. Patients initiated on clindamycin should be monitored closely and escalated to TMP/SMX or doxycycline if improvement is not demonstrated within 24-48 hours of therapy initiation.

Table 2: Treatment of Pyomyositis

Pyomyositis		
Classification	Comment	Recommendations and Treatment
Pyomyositis	- Consider surgery consult for debridement and/or drainage - Consider ID consultation - Recommend obtaining 2 sets of blood cultures - MRI is the recommended imaging modality for diagnosis of pyomyositis. - Early drainage of purulent material is recommended Treatment duration: 14-21 days	Empiric therapy: IV vancomycin^{*(a)} - In patients with preceding trauma, consider treating similar to necrotizing fasciitis. Defined therapy (IV)^(b) <i>S. aureus</i> (MRSA): IV vancomycin^{*(a)} <i>S. aureus</i> (MSSA): - IV nafcillin 200 mg/kg/day divided Q4h or as CIVI (max: 12 g/DAY) <u>OR</u> - IV cefazolin 33 mg/kg/dose Q8h (max dose: 2 g) <i>S. pyogenes</i> (GAS): - IV aqueous penicillin G^(c) 50,000 units/kg/dose Q4h (max dose: 4 million units) - Consider the addition of IV clindamycin^(d) 13 mg/kg/dose Q8h (max dose: 900 mg) in patients with hemodynamic instability

CIVI: continuous IV infusion; *Vancomycin dosed per pediatric IV vancomycin order set [PED-Vancomycin IV] or pharmacy to dose; **(a)**: Other IV agents active against MRSA include daptomycin or linezolid; **(b)**: When initial IV therapy used, consider conversion to PO after favorable clinical response observed and bacteremia ruled out; **(c)**: Parenteral aqueous penicillin G is available in sodium and potassium salt forms. Due to differences in cost, the potassium formulation of penicillin G is generally preferred; in patients with hyperkalemia, may consider utilization of the sodium formulation; **(d)**: Clindamycin is added for anti-toxin effect associated with *S. pyogenes* (GAS) and may be discontinued after 72h if blood cultures cleared and patient clinically improved.

Table 3: Treatment of Abscesses, Furuncles, and Carbuncles

Abscess, Furuncles, and Carbuncles		
Severity	Comment	Treatment
Mild No signs of systemic infection ^(a)	- Incision and drainage - Mark the border of erythema and follow-up in 48-72 hours to assure response	<u>Systemic antibiotics often not required</u> ^(11, 12) if there is no overlying cellulitis and the patient is immunocompetent - For small or already draining collections, the use of warm compresses alone may be indicated -- <i>May consider up to 5 days of PO antibiotics (see moderate section for guidance on agent selection and dosing) if culture(s) positive for MRSA or if there is surrounding cellulitis</i>
Moderate Patients with signs of systemic infection ^(a)	- Incision and drainage - Consider obtaining one set of blood cultures[¶] Treatment duration: 5 days	- PO TMP-SMX^Φ 6 mg/kg/dose BID (<i>based on trimethoprim component</i>) (max dose: 320 mg trimethoprim) <u>OR</u> - PO doxycycline^Φ 2.2 mg/kg/dose BID (max dose: 100 mg) <u>OR</u> - PO clindamycin^(h) 10 mg/kg/dose TID (max dose: 450 mg) -- <i>KCH inducible clindamycin resistance rates are >20% for MSSA and >30% for MRSA</i>
Severe ^(b) Patients who previously failed I&D <u>plus</u> oral antibiotics, immunocompromised patients, or patients with signs of systemic infection ^(a)	- Consider surgery consult for incision and debridement - Consider ID consultation - Recommend obtaining at least one set of blood cultures; may consider a second set of blood cultures if able[¶] - IV therapy warranted initially with prompt de-escalation to PO as appropriate^(d) Treatment duration (abscess): 5-14 days from last debridement	Empiric therapy: IV vancomycin^{*(c)} Defined therapy (IV)^(d) <i>S. aureus</i> (MRSA): IV vancomycin^{*(e)} <i>S. aureus</i> (MSSA): - IV nafcillin 200 mg/kg/day divided Q4h or as CIVI (max: 12g/DAY) <u>OR</u> - IV cefazolin 33 mg/kg/dose Q8h (max dose: 2 g) <i>S. pyogenes</i> (GAS): - IV aqueous penicillin G^(d) 50,000 units/kg/dose Q4h (max dose: 4 million units) • <i>Consider the addition of IV clindamycin^(e) 13 mg/kg/dose Q8h (max dose: 900 mg) in patients with hemodynamic instability</i> Defined therapy (PO)^(f) <i>S. aureus</i> (MRSA): see antibiotics above in “moderate” section <i>S. aureus</i> (MSSA): - PO cefadroxil 15 mg/kg/dose PO BID (max dose: 500 mg) <u>OR</u> - PO cephalexin 25 mg/kg/dose QID (max dose: 1000 mg) <i>S. pyogenes</i> (GAS): - PO Penicillin VK 15-20 mg/kg/dose PO QID (max dose: 500 mg) <u>OR</u> - PO cefadroxil 15 mg/kg/dose PO BID (max dose: 500 mg) Type I Penicillin allergy^(g): PO Clindamycin^(h) 10 mg/kg/dose TID (max dose: 450 mg)

CIVI: continuous IV infusion; *Vancomycin dosed per pediatric IV vancomycin order set [PED-Vancomycin IV] or pharmacy to dose; ¶: If blood cultures obtained at outside hospital, consider redrawing blood culture upon arrival to UK; Φ: Although *S. aureus* represents the most commonly implicated pathogen in purulent SSTIs, the addition of PO penicillin VK or PO amoxicillin may be considered in patients who are not improving despite drainage and therapy with doxycycline or TMP/SMX, as GAS resistance rates are estimated to be as high as 20% with doxycycline and 10% with TMP/SMX; (a): Signs of systemic infection include: fever (>38°C), tachycardia, tachypnea, leukocytosis, or leukopenia; (b): Consider Infectious Disease consultation in patients with bacteremia, anticipation of greater than 2 weeks of anti-infective treatment, immunocompromising conditions, history of multi-drug resistant organisms, or recurrence of infection; (c): Other IV agents active against MRSA include daptomycin or linezolid; (d): Parenteral aqueous penicillin G is available in sodium and potassium salt forms. Due to differences in cost, the potassium formulation of penicillin G is generally preferred; in patients with hyperkalemia, may consider utilization of the sodium formulation. (e): Clindamycin is added for anti-toxin effect associated with *S. pyogenes* (GAS) and may be discontinued after 72h if blood cultures cleared and patient clinically improved; (f): When initial IV therapy used, consider conversion to PO after favorable clinical response observed and bacteremia ruled out; (g): Type I reaction includes anaphylaxis, angioedema, or urticaria; in patients with minor penicillin allergies (non-specific rashes, intolerance, etc.), cross reactivity with cephalosporins is low (<3%)(1, 2) (h): Clindamycin susceptibility rates per the 2021/2022 KCH antibiogram are only 69% for MRSA and 79% for MSSA. Patients initiated on clindamycin should be monitored closely and escalated to TMP/SMX or doxycycline if improvement is not demonstrated within 24-48 hours of therapy initiation.

Table 4: Treatment of Necrotizing Fasciitis, Clostridial Gas Gangrene, or Myonecrosis^(3, 7)

Necrotizing Fasciitis, Clostridial Gas Gangrene, or Myonecrosis	
<u>EMERGENT surgical debridement is warranted.</u> Obtain deep tissue culture and at least one set of blood cultures[¶]. If >20% BSA involvement, consider wound care and burn unit referral. <u>IMMEDIATE surgical consultation necessary</u> <u>ID consultation strongly recommended</u> Definition: Aggressive infection marked by extensive tissue damage and necrosis that tracks along the fascial plane overlying muscle. Clinical features include rapid progression, gas/crepitus, severe pain/with passive stretch, and systemic signs of infection and/or organ damage. This condition is associated with high mortality and is a <u>clinical diagnosis</u> (if suspicion, emergent surgical consultation is required for debridement prior to or in conjunction with imaging). Treatment duration: Antimicrobial therapy should be administered until 48 hours following last debridement. A shorter total course (<7 days) appears to be equivalent to longer courses of therapy and should be considered.⁽³⁾	
Empiric regimen: The utilization of linezolid, in place of vancomycin, provides the following benefits: - Excellent coverage of Gram-positive organisms of concern - Reduced risk of nephrotoxicity <i>Anti-toxin effect</i>	Preferred: - IV/PO linezolid 10 mg/kg/dose (Q8h if <12 years, Q12h if ≥12 years) (max dose: 600 mg) <u>PLUS</u> - IV piperacillin/tazobactam^(a) 100 mg/kg/dose (<i>based on piperacillin component</i>) Q6h (max dose: 4.5 g) Alternative: - IV vancomycin* <u>PLUS</u> - IV cefepime 50 mg/kg/dose Q8h (max dose: 2 g) <u>PLUS</u> - IV/PO metronidazole 10 mg/kg/dose Q8h (max dose: 500 mg) <u>PLUS</u> - IV clindamycin^(b) 13 mg/kg/dose Q8h (max dose: 900 mg)
Pathogen-specific regimen:	Type I (mixed aerobic and anaerobic flora): De-escalate therapy based on culture data Type II (monomicrobial): <i>Clostridium</i> species: - IV aqueous penicillin G^(c) 50,000 units/kg/dose Q4h (max dose: 4 million units) <u>PLUS</u> - IV clindamycin^(b) 13 mg/kg/dose Q8h (max dose: 900 mg) <i>S. pyogenes</i>: - IV aqueous penicillin G 50,000 units/kg/dose Q4h (max dose: 4 million units) <u>PLUS</u> - IV clindamycin^(b) 13 mg/kg/dose Q8h (max dose: 900 mg) <i>S. aureus</i> (MSSA): - IV cefazolin 33 mg/kg/dose Q8h (max dose: 2000 mg) <u>OR</u> - IV nafcillin 200 mg/kg/day divided Q4h (max: 12g/DAY) <i>S. aureus</i> (MRSA): - IV/PO linezolid 10 mg/kg/dose (max dose: 600 mg) - Q8h if <12 years - Q12h if ≥12 years <u>OR</u> - IV vancomycin* <i>Vibrio vulnificus</i> or <i>Aeromonas hydrophila</i>^(d): - IV/PO doxycycline^(e) 2.2 mg/kg/dose Q12h (max dose: 100 mg) <u>PLUS</u> - IV/IM ceftriaxone 50 mg/kg/dose Q24h (max dose: 2000 mg)

¶: If blood cultures obtained at outside hospital, consider redrawing blood culture upon arrival to UK; *Vancomycin dosed per pediatric IV vancomycin order set [PED-Vancomycin IV] or pharmacy to dose; (a): Piperacillin/tazobactam dosing is based on piperacillin component. Piperacillin/tazobactam 4.5 g contains 4000 mg piperacillin and 500 mg tazobactam; (b): Clindamycin is added for anti-toxin effect associated with *S. pyogenes* (GAS) and may be discontinued after 72h if blood cultures cleared and patient clinically improved; (c): Parenteral aqueous penicillin G is available in sodium and potassium salt forms. Due to differences in cost, the potassium formulation of penicillin G is generally preferred; in patients with hyperkalemia, may consider utilization of the sodium formulation; (d): *Vibrio* and *Aeromonas* should be considered in patients with wound exposure to fresh or brackish water; (e): Original data describing enamel staining are from studies with tetracycline. Recent comparative data in children suggest that doxycycline is NOT likely to cause visible teeth staining or enamel hypoplasia in children. The American Academy of Pediatrics recommend that doxycycline can be administered in any pediatric patient for durations ≤21 days without regard to the patient's age.⁽⁴⁾

Table 5: Treatment of Toxic Shock Syndrome or Staphylococcal Scalded Skin Syndrome^(3, 7)

Toxic Shock Syndrome or Staphylococcal Scalded Skin Syndrome	
Obtain at least one set of blood cultures; may consider a second set of blood cultures if able [¶]	
Consider ID consultation Rationale for empiric therapy includes coverage of <i>S. aureus</i> (including MRSA) and <i>Streptococcus sp.</i> with the addition of a ribosomal-active agent (ie, clindamycin or linezolid) for anti-toxin effect in the first 48-72 hours of infection and until patient demonstrates blood culture clearance and clinical improvement.	
Empiric regimen:	- IV nafcillin 33 mg/kg/dose Q4h (max dose: 2000 mg) <u>PLUS</u> - IV/PO linezolid 10 mg/kg/dose (Q8h if <12 years, Q12h if ≥12 years) (max dose: 600 mg)
Pathogen-specific regimen:	<i>S. pyogenes</i>: - IV aqueous penicillin G^(a) 50,000 units/kg/dose Q4h (max dose: 4 million units) <u>PLUS</u> - IV clindamycin^(b) 13 mg/kg/dose Q8h (max dose: 900 mg) <i>S. aureus</i> (MSSA): - IV cefazolin 33 mg/kg/dose Q8h (max dose: 2000 mg) <u>PLUS</u> - IV clindamycin^(b) 13 mg/kg/dose Q8h (max dose: 900 mg) <u>OR</u> - IV nafcillin 200 mg/kg/day divided Q4h (max: 12g/DAY) <u>PLUS</u> - IV clindamycin^(b) 13 mg/kg/dose Q8h (max dose: 900 mg) <i>S. aureus</i> (MRSA): - IV vancomycin* <u>PLUS</u> - IV clindamycin^(b) 13 mg/kg/dose Q8h (max dose: 900 mg) <u>OR</u> - IV/PO linezolid 10 mg/kg/dose (Q8h if <12 years, Q12h if ≥12 years) (max dose: 600 mg)

*Vancomycin dosed per pediatric IV vancomycin order set [PED-Vancomycin IV] or pharmacy to dose; **(a)**: Parenteral aqueous penicillin G is available in sodium and potassium salt forms. Due to differences in cost, the potassium formulation of penicillin G is generally preferred; in patients with hyperkalemia, may consider utilization of the sodium formulation; **(b)**: Clindamycin is added for anti-toxin effect associated with *S. pyogenes* (GAS) and may be discontinued after 48-72h if blood cultures cleared and patient clinically improved; **¶**: If blood cultures obtained at outside hospital, consider redrawing blood culture upon arrival to UK

Table 6: Preemptive Therapy (Prophylaxis) and Treatment of Human, Cat, or Dog Bites

PRE-EMPTIVE Therapy (ie, Prophylaxis) Following Human, Cat, or Dog Bites ²	
See Recommendations for Tetanus (p. 131) and Rabies (p. 177) in the UK Pediatric Trauma Manual Pre-emptive antimicrobial therapy is recommended in patients who present <u>within 24 hours</u> of bite and who meet one or more of the following criteria: • Patients who are immunocompromised, asplenic, or who have advanced liver disease • Patients who have edema of the affected area • Patients who have moderate to severe injuries, especially to the hands or face • Patients who have injuries that may have penetrated the periosteum or joint capsule Organisms of Concern: <i>S. aureus</i>, <i>Streptococcus sp.</i>, <i>Pasteurella multocida</i>, <i>Capnocytophaga canimorsus</i>, and oral anaerobes	
Preferred <u>pre-emptive</u> regimen	• PO amoxicillin/clavulanate 20 mg/kg/dose amoxicillin (<i>dosing based on amoxicillin component</i>) q12h (max dose: 875 mg amoxicillin) • Total treatment duration: 3 days
<u>Pre-emptive</u> regimen if anaphylactic penicillin allergy	• IV/PO doxycycline^(a) 2.2 mg/kg/dose BID (max dose: 100 mg) <i>regardless of age</i> • Total treatment duration: 3 days
Treatment of Infected Human, Cat, or Dog Bites	
For complicated or severe wounds, ID consultation recommended Organisms of Concern (cat or dog): <i>S. aureus</i>, <i>Streptococcus sp.</i>, <i>Pasteurella multocida</i>, <i>Capnocytophaga canimorsus</i>, and oral anaerobes Organisms of Concern (human): <i>S. aureus</i>, <i>Streptococcus sp.</i>, <i>Eikenella corrodens</i>, and oral anaerobes	
Preferred treatment regimens	• PO amoxicillin/clavulanate 20 mg/kg/dose amoxicillin (<i>dosing based on amoxicillin component</i>) Q8h (max dose: 875 mg amoxicillin) • OR • IV ampicillin/sulbactam 50 mg/kg/dose ampicillin (<i>dosing based on ampicillin component</i>) Q6h (max dose^(b): 3 g) • Total treatment duration: 5 days
Treatment regimen if anaphylactic penicillin allergy	• IV/PO doxycycline^(a) 2.2 mg/kg/dose BID (max dose: 100 mg) • Total treatment duration: 5 days

(a): Original data describing enamel staining are from studies with tetracycline. Recent comparative data in children suggest that doxycycline is NOT likely to cause visible teeth staining or enamel hypoplasia in children. The American Academy of Pediatrics have revised previous warnings and recommend that doxycycline can be administered in any pediatric patient for durations ≤21 days without regard to the patient's age; (b): Ampicillin/sulbactam dosing is based on ampicillin component; Ampicillin/sulbactam 3 g contains 2 g ampicillin and 1 g sulbactam.

Table 7: Treatment and Management of Preseptal and Orbital Cellulitis

Preseptal and Orbital Cellulitis	
CONSULTATION: Preseptal Cellulitis: Infection anterior to the orbital septum - Consider ophthalmology consult and ID consult as clinically indicated Orbital Cellulitis: Infection posterior to the orbital septum - Ophthalmology consult and ID consult strongly recommended for all cases of orbital cellulitis - Neurosurgery consult should be considered in cases of orbital cellulitis with intracranial extension - ENT consult should be considered in cases of orbital cellulitis with significant sinus disease - OMFS consult should be considered in cases of orbital cellulitis with possible dental source	
DIAGNOSTICS: Obtain MRSA nares PCR and at least one set of blood cultures in all patients with orbital cellulitis ^(13, 14)	
IMAGING: - A contrasted CT of orbits and sinuses is strongly recommended in any of the following cases: - Presence of one or more orbital symptoms (See Table 8) - Unable to fully examine eye (ie swollen shut) or patient unable to verbalize symptoms - Symptoms of significant headache or nausea/vomiting - Failure of outpatient treatment with oral antibiotics - Contrasted MRI of head is strongly recommended in patients with concern for intracranial extension on contrasted CT of orbits/sinuses - Notably, the clinical features for both preseptal and orbital cellulitis are not always well demarcated^(a) and there may be considerable overlap in signs/symptoms. Contrasted CT may be considered in patients with unclear symptoms.^(13,14)	
MICROBIOLOGY: - Most commonly (up to 90%) develop secondary to sinusitis: <i>S. pneumoniae</i>, <i>H. influenzae</i>, <i>S. aureus</i>, oral anaerobes - May also arise secondary to penetrating trauma, surgery, endophthalmitis, or dental surgery: <i>S. aureus</i>, <i>S. pyogenes</i> - Presence of subperiosteal or orbital abscess: <i>Streptococcus sp.</i>, <i>H. influenzae</i>, <i>S. aureus</i>, Gram-negative bacilli and anaerobes; more frequently monomicrobial in children <9 years of age; predominantly polymicrobial in children ≥9 years of age	
Definitions	Empiric Therapy
Preseptal^Φ Cellulitis <u>with clear skin source</u>	- IV vancomycin* OR - PO TMP/SMX 6 mg/kg/dose BID (max dose: 320 mg TMP) OR - PO doxycycline^(f) 2.2 mg/kg/dose BID (max dose: 100 mg) Total treatment duration (IV and PO): 7 days
Preseptal^Φ Cellulitis <u>with sinusitis, dental, or otherwise unclear source</u>	- IV ampicillin/sulbactam^(b) 75 mg/kg/dose (<i>dosing based on ampicillin component</i>) Q6h (max dose: 3 g) OR - PO amoxicillin/clavulanate* 30 mg/kg/dose (<i>dosing based on amoxicillin component</i>) TID (max dose: 1200 mg) Total treatment duration (IV and PO): 7 days <i>If type I penicillin allergy^(b): PO doxycycline 2.2 mg/kg/dose BID (max dose: 100 mg)</i>
Orbital Cellulitis	NO INTRACRANIAL EXTENSION (FOLLOWING CONTRASTED CT +/- MRI): - IV ampicillin/sulbactam^(c) 75 mg/kg/dose ampicillin (<i>dosing based on ampicillin component</i>) Q6h (max dose: 3 g) - Discuss timing of de-escalation to oral therapy with ID (typically following ≥72 hours of IV therapy) Total treatment duration (IV + PO): 10 days
	INTRACRANIAL EXTENSION OR PRIOR TO CONTRASTED CT OR MRI: - IV ceftriaxone 50 mg/kg/dose Q12h (max dose: 2000 mg) PLUS - PO/IV metronidazole 10 mg/kg/dose Q8h (max dose: 500 mg) PLUS - IV vancomycin* Total treatment duration: ID consult
	SUBPERIOSTEAL OR ORBITAL ABSCESS (+/- combination with surgical drainage as guided by ophthalmology): - IV ceftriaxone 50 mg/kg/dose Q12h (max dose: 2000 mg) PLUS - PO/IV metronidazole 10 mg/kg/dose Q8h (max dose: 500 mg) PLUS - IV vancomycin* Total treatment duration: ID consult; discuss possibility of de-escalation to oral therapy with ID

*Vancomycin dosed per pediatric IV vancomycin order set [PED-Vancomycin IV] or pharmacy to dose; ^Φ: The term periorbital cellulitis, sometimes used interchangeably with preseptal cellulitis, should be avoided as it does not make the distinction between orbital and preseptal cellulitis clear; [†]: For patients ≥40 kg who are able to swallow tablets, may consider amoxicillin/clavulanate XR tablets at a dose of 2000 mg BID (often requires prior authorization) (a): As orbital signs may be difficult to evaluate in pediatric patients with orbital cellulitis^(13, 14); (b): Type I reaction includes anaphylaxis, angioedema, or urticaria (c): Ampicillin/sulbactam dosing is based on ampicillin component; Ampicillin/sulbactam 3 g contains 2 g ampicillin and 1 g sulbactam.

Table 8: Definitions and Pathogens of Concern

Impetigo, Folliculitis, and Ecthyma	
Definitions	Common Pathogens
Impetigo: Lesions begin as erythematous papules; vesicles rupture and discharge forms honey-colored crusts on an erythematous base; may be bullous or non-bullous	Bullous: <i>Staphylococcus aureus</i> , typically MSSA Non-bullous: <i>Staphylococcus aureus</i> including MRSA, <i>Streptococcus pyogenes</i> (GAS)
Folliculitis: Infections of the hair follicle; superficial inflammation and purulence limited to the epidermis	<i>Staphylococcus aureus</i> <i>Pseudomonas aeruginosa</i> (hot tub)
Ecthyma: Superficial dermal infection, circular, erythematous ulcers with adherent crust, often with surrounding erythematous edema	<i>Staphylococcus aureus</i> , <i>Streptococcus pyogenes</i> (GAS)
Cellulitis and Erysipelas	
Definitions	Common Pathogens
Cellulitis: Area of skin with erythema, edema, and warmth; extends to deep subcutaneous tissue; less distinct borders	<i>β-hemolytic Streptococcus</i> [<i>S. dysgalactiae</i> , <i>S. pyogenes</i> (GAS), <i>S. agalactiae</i> (GBS)], rarely <i>Staphylococcus aureus</i>
Erysipelas: Erythematous, indurated plaque with clearly demarcated border; involves upper dermis; more superficial and raised than cellulitis, acute onset typically with a rash on legs, toes, face, arms, and/or fingers	
Necrotizing Fasciitis	
Definitions	Common Pathogens
Necrotizing fasciitis: Aggressive infection marked by extensive tissue damage and necrosis that tracks along the fascial plane overlying muscle. Clinical features include rapid progression, gas/crepitus, severe pain/with passive stretch, and systemic signs of infection and/or organ damage. This condition is associated with high mortality and is a <u>clinical diagnosis</u> (if suspicion, emergent surgical consultation is required for debridement prior to or in conjunction with imaging). • <u>Fournier's gangrene:</u> infective necrotizing fasciitis affecting the external genitalia, perineal or perianal regions • <u>Ludwig's angina:</u> severe diffuse necrotizing cellulitis that presents with acute onset and spreads rapidly, bilaterally affecting the submandibular, sublingual and submental spaces • <u>Omphalitis:</u> Infection of the umbilicus and surrounding tissue which occurs primarily in the neonate.	
Type I: Polymicrobial infection; most commonly associated with perineal abscess, penetrating abdominal trauma, surgical procedures involving the bowel, decubitus ulcers, or injection sites in persons who inject drugs. In this scenario, an average of 5 pathogens isolated from culture(s).	Anaerobes, staphylococci, streptococci, Gram-negative organisms
Type II: Monomicrobial infection; typically occurs in the limbs	<i>Streptococcus pyogenes</i> (GAS), <i>Staphylococcus aureus</i> including MRSA, and anaerobic streptococci Water-related: <i>Vibrio vulnificus</i> , <i>Aeromonas hydrophila</i>
Clostridial Gas Gangrene or Myonecrosis: Infection involving deeper tissue such as a muscle, leading to a rapidly spreading infection along tissue planes. Usually secondary to trauma but also can arise spontaneously.	<i>Clostridium perfringens</i> , <i>Clostridium novyi</i> , <i>Clostridium histolyticum</i> , <i>Clostridium septicum</i>
Omphalitis: Infection of the umbilicus and/or surrounding tissues. Predominantly in neonates.	<i>Streptococcus pyogenes</i> (GAS), <i>Staphylococcus aureus</i> including MRSA, <i>E. coli</i> , <i>K. pneumoniae</i> , and <i>Clostridium spp.</i>
Abscesses, Furuncles, and Carbuncles	
Definitions	Common Pathogens
Abscess: Collection of pus within the dermis and deeper skin tissues, clinically manifests as painful, tender, fluctuant, and erythematous nodules.	<i>Staphylococcus aureus</i> , including community-acquired MRSA, <i>Streptococcus pyogenes</i> (GAS)
Furuncles (boils): Infection of hair follicles; purulence extends into subcutaneous tissue, creating a small abscess.	
Carbuncle: Infection of multiple adjacent follicles, frequently localized on the back of the neck, shoulders, or thighs; usually has one or more openings that drain purulence onto the skin.	

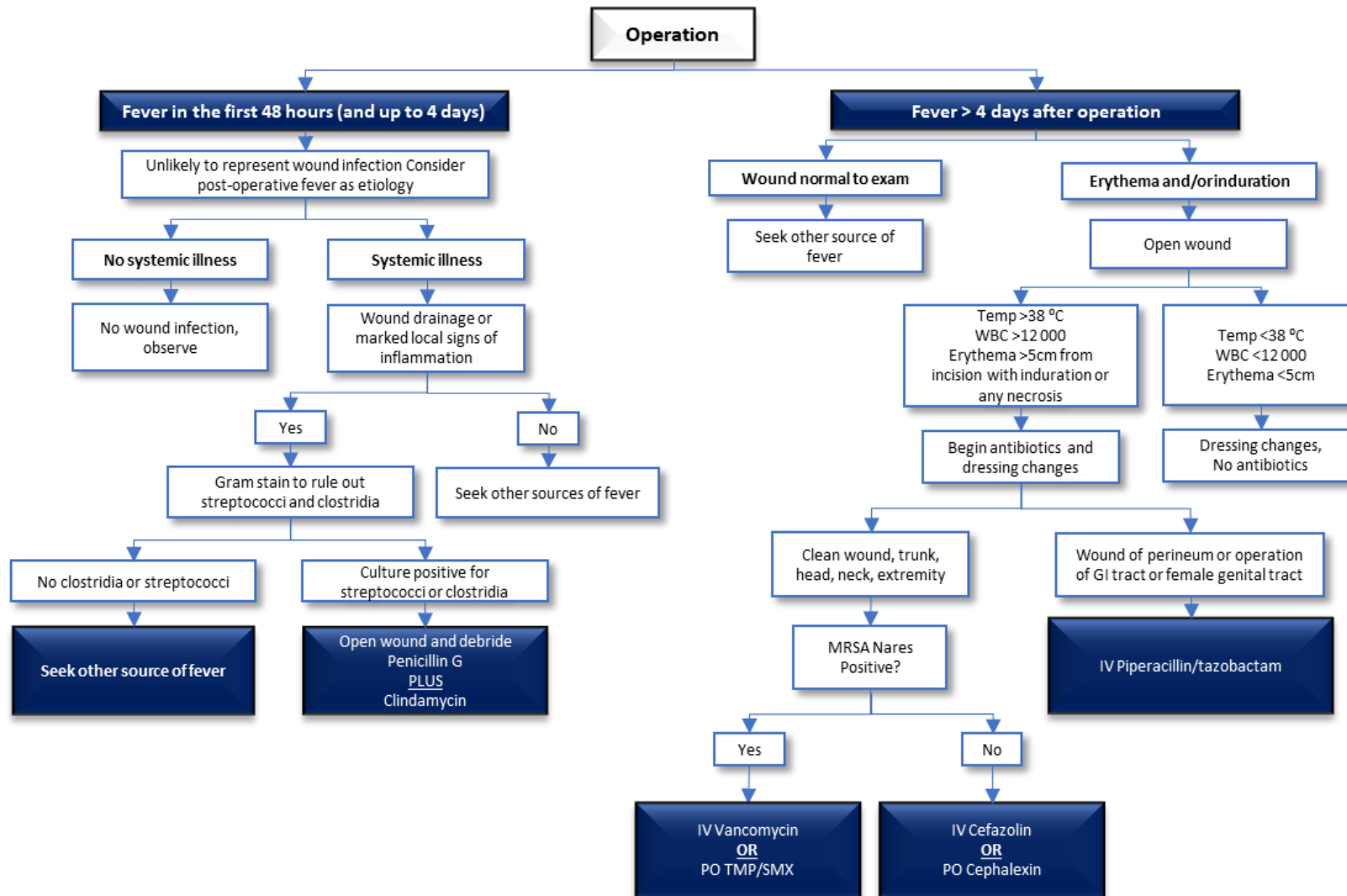
Abbreviations: MRSA, methicillin-resistant *Staphylococcus aureus*; MSSA, methicillin-susceptible *Staphylococcus aureus*

Superficial/ Deep Incisional Surgical Site Infections

Suture removal plus incision and drainage should be performed for surgical site infections

- Adjunctive antibiotics are not routinely recommended
- Empiric antibiotic treatment should be started in the following cases:
 - Signs of systemic infection
 - Erythema and induration extending >5cm from the wound edge

Figure 5: Algorithm for the Management and Treatment of Surgical Site Infections



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